

Kevin Maa

kmaa302@gmail.com | [linkedin.com/in/kevinmaa](https://www.linkedin.com/in/kevinmaa) | github.com/dz337

EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Information Technology with Honors | GPA: 3.62

May 2027

Minor: Secure Comp & Networks | *Honors:* Burnett Honors College – University Honors

Relevant Courses: Computer Architecture Concepts, Network Security & Privacy, Security in Computing

Affiliations: IEEE UCF

PROJECTS

KnightCore – FPGA Integration – 2025 AMD Hardware Competition – Group Project Apr 2025 – Aug 2025

- Solely responsible for integrating custom GPU hardware module onto Red Pitaya Zynq FPGA platform using Vivado block design, connecting Zynq processor, AXI interconnect, BRAM controller/memory, configuring address spaces in the housekeeping region, and generating bitstreams, enabling team to focus exclusively on hardware design.
- Debugged memory-mapped I/O communication issues between ARM processor and FPGA programmable logic, identifying and correcting address mapping errors in shared memory regions to enable successful hardware integration.
- Developed and executed shell script and Python-based validation tests on Red Pitaya to verify register read/write operations and confirm successful processor-to-FPGA communication through memory-mapped interfaces.

CoreRV – Personal RISC-V SoC – Solo Project Summer 2025 – Present

- Designing a custom RV32IMA soft-core SoC in SystemVerilog targeting the Red Pitaya Stemlab 125-14 FPGA, with the goal of running Doom bare-metal, built around a 5-stage pipeline with full hazard detection, data forwarding, and load-use stall logic.
- Implementing a multi-cycle iterative hardware divider and 3-cycle pipelined multiplier with dedicated FSMs and pipeline stall integration, alongside a Machine-mode CSR subsystem supporting precise trap handling, Writeback-stage exception delivery, and MRET return-from-trap.
- Actively adding dynamic branch prediction, L1 instruction and data caches, UART, a memory-mapped timer, SPI flash, and a framebuffer with HDMI output to complete the SoC peripheral stack.

ESP32 Motion Gesture Classifier – Solo Project Mar 2026 – Present

- Building an embedded gesture recognition system on an ESP32 DevKit V1 wired to an MPU-6050 6-axis IMU over I2C, writing C++ firmware to stream raw accelerometer and gyroscope data at 77Hz for data collection.
- Training a deep learning classifier in Edge Impulse using Spectral Analysis and Fast Fourier Transform feature extraction to distinguish four motion classes: idle, wave, circle, and thrust, achieving high accuracy on-device.
- Deploying the trained model as a C++ library back onto the ESP32 for fully offline real-time inference, completing a full TinyML pipeline from raw sensor data to edge deployment without cloud dependency.

Self-Hosted Workstation Server – Solo Project Sep 2025 – Present

- Administering a Docker-based self-hosted stack (Gitea, Nextcloud, Immich, Paperless-ngx, Vaultwarden, Ollama) on a 32-core/64-thread Linux Mint workstation alongside a suite of VirtualBox VMs for isolated lab environments.
- Deploying a WireGuard-based Tailscale mesh with a self-hosted AdGuard Home DNS resolver, filtering ads and trackers network-wide via layered blocklists including OISD and HaGeZi Threat Intelligence.
- Hardening credential hygiene by locking down config file permissions and migrating password/TOTP management to a self-hosted Vaultwarden instance for one-click autofill.

LEADERSHIP

IEEE-Eta Kappa Nu Zeta Chi – Treasurer May 2026 – Present

Electrical and Computer Engineering Honor Society, University of Central Florida

- Inducted into HKN as one of the top-performing Information Technology students meeting the chapter GPA threshold, joining a selective cohort of ECE and CPE majors recognized for academic excellence.
- Secured chapter funding by submitting a CRT request to UCF Student Government, covering registration costs for members to attend the 2026 IEEE-HKN Student Leadership Conference.

TECHNICAL SKILLS

Languages: SystemVerilog, C, C++, Python, Bash, Assembly (RISC-V), TypeScript, JavaScript, HTML, CSS

RTL and Hardware Design: RISC-V ISA, Pipelined CPU Design, AXI Interconnect, Memory-Mapped I/O, BRAM, CORDIC, UART, Hazard Detection and Forwarding

Embedded and Edge AI: ESP32, Raspberry Pi, MPU-6050, I2C, Edge Impulse, TinyML

Virtualization and Infrastructure: VirtualBox, Tailscale/WireGuard Mesh VPN, Self-Hosted Docker Services, DNS Filtering (AdGuard Home)

Network Security: Wireshark, Security Onion, Kali Linux, OPNsense

Tools: Git, GitHub, Linux, Windows, VSCode, Vivado, Docker, VLAN